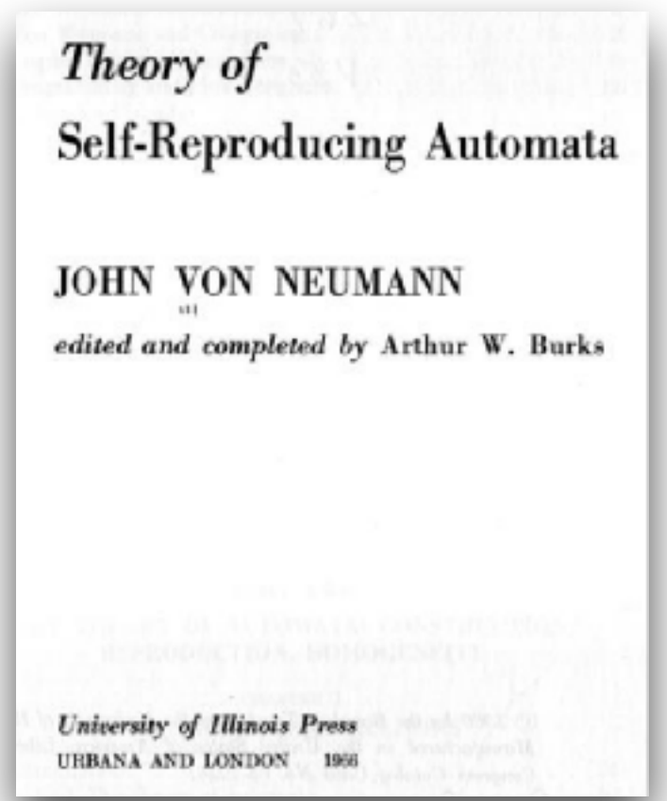
From Conway to LangGraph: Agent Systems for Physicists

From Von Neumann's Dreams to Wolfram's Complexity

The dawn of computing in physics (1940s-1950s)

Von Neumann (1940s):

- Self-replicating automata
- Motivation: understanding the logic of life
- Problem: models too complex (29 states!)



Theory of
Self-Reproducing Automata

JOHN VON NEUMANN

edited and completed by Arthur W. Burks

University of Illinois Press
URBANA AND LONDON 1966

From Wikipedia, the free encyclopedia

The native form of this personal name is Neumann János Lajos. *This article uses Western name order when mentioning individuals.*

John von Neumann (/von ˈnaɪmən/ *von NOY-mən*; Hungarian: *Neumann János Lajos* [ˈnɔjmɔn ˈjaːnoʃ ˈlɒjoʃ]; December 28, 1903 – February 8, 1957) was a Hungarian and American mathematician, physicist, computer scientist and engineer. Von Neumann had perhaps the widest coverage of any mathematician of his time,[9] integrating pure and applied sciences and making major contributions to many fields, including mathematics, physics, economics, computing, and statistics. He was a pioneer in building the mathematical framework of quantum physics, in the development of functional analysis, and in game theory, introducing or codifying concepts including cellular automata, the universal constructor and the digital computer. His analysis of the structure of self-replication preceded the discovery of the structure of DNA.

During World War II, von Neumann worked on the Manhattan Project. He developed the mathematical models behind the explosive lenses used in the implosion-type nuclear weapon.[10] Before and after the war, he consulted for many organizations including the Office of Scientific Research and Development, the Army's Ballistic Research Laboratory, the Armed Forces Special Weapons Project and the Oak Ridge National Laboratory.[11] At the peak of his influence in the 1950s, he chaired a number of Defense Department committees including the Strategic Missile Evaluation Committee and the ICBM Scientific Advisory Committee. He was also a member of the influential Atomic Energy Commission in charge of all atomic energy development in the country. He played a key role alongside Bernard Schriever and Trevor Gardner in the design and development of the United States' first ICBM programs.[12] At that time he was considered the nation's foremost expert on nuclear weaponry and the leading defense scientist at the U.S. Department of Defense.

Von Neumann's contributions and intellectual ability drew praise from colleagues in physics, mathematics, and beyond. His accolades include a Medal of Freedom and a crater on the Moon named in his honor.

Life and education [edit]

Family background [edit]

Von Neumann was born in Budapest, Kingdom of Hungary (then part of Austria-Hungary),[13][14][15] on December 28, 1903, to a wealthy, non-observant Jewish family.

John von Neumann

von Neumann in the 1940s

Member of the United States Atomic Energy Commission	
In office	March 15, 1955 – February 8, 1957
President	Dwight D. Eisenhower
Preceded by	Eugene M. Zuckert
Succeeded by	John S. Graham
Personal details	
Born	Neumann János Lajos December 28, 1903 Budapest, Kingdom of Hungary, Austria-Hungary
Died	February 8, 1957 (aged 53) Washington, D.C., U.S.
Resting place	Princeton Cemetery
Citizenship	Hungary United States
Education	Pázmány Péter University University of Berlin ETH Zürich

Scientific career

ENIAC (Electronic Numerical Integrator and Computer)

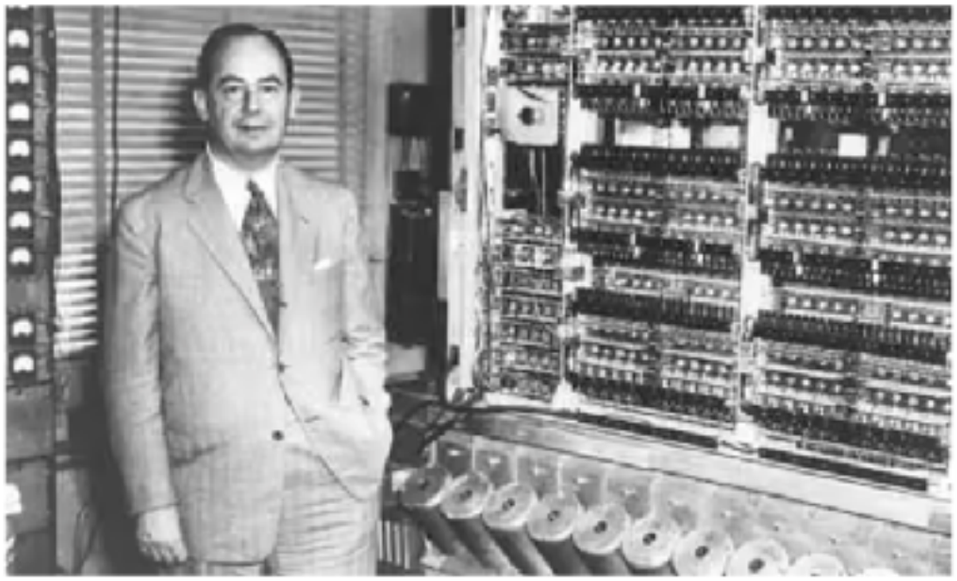
was designed and built by:

- **J. Presper Eckert**
- **John W. Mauchly**

University of Pennsylvania
Construction: 1943–1945
Public unveiling: February 1946

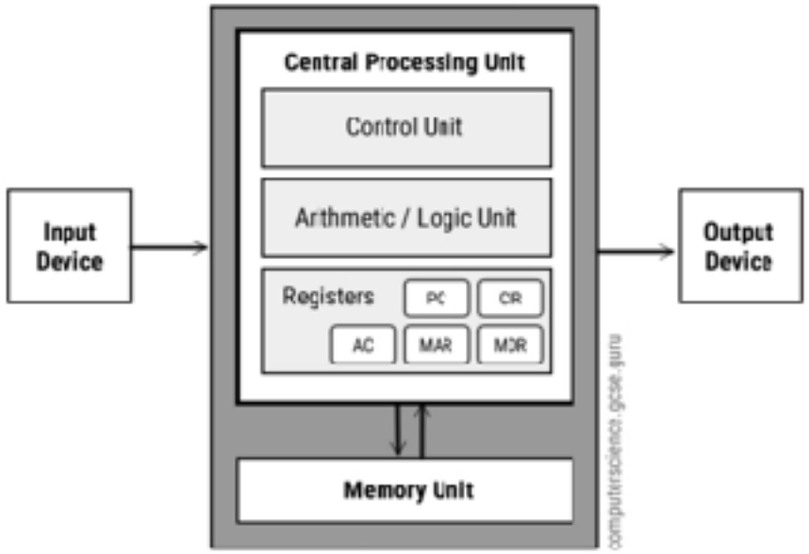
It was originally funded by the U.S. Army for ballistic calculations.

The MANIAC……

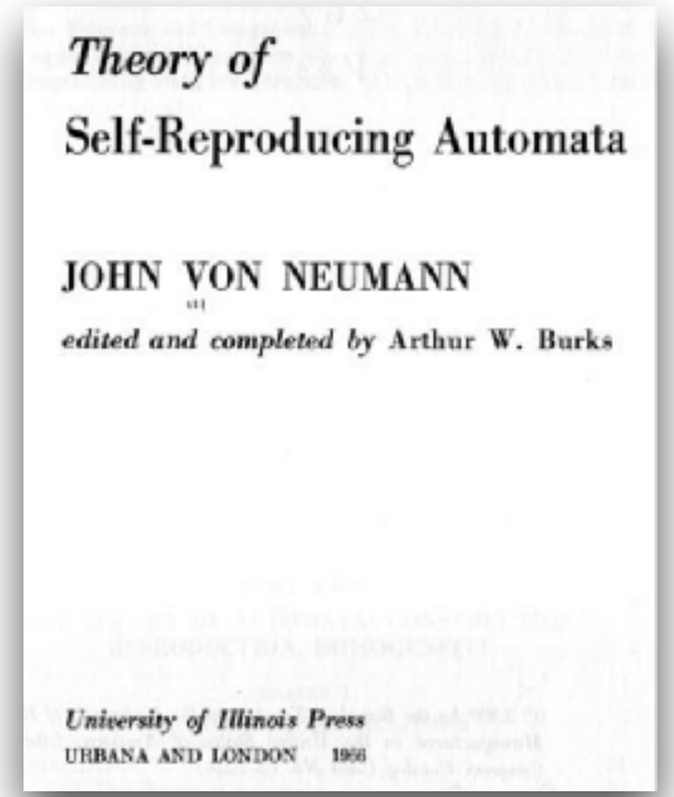


John von Neumann with the stored-program computer at the Institute for Advanced Study, Princeton, New Jersey, in 1945. Photograph: Getty

The von Neumann architecture



The dawn of computing in physics (1940s-1950s)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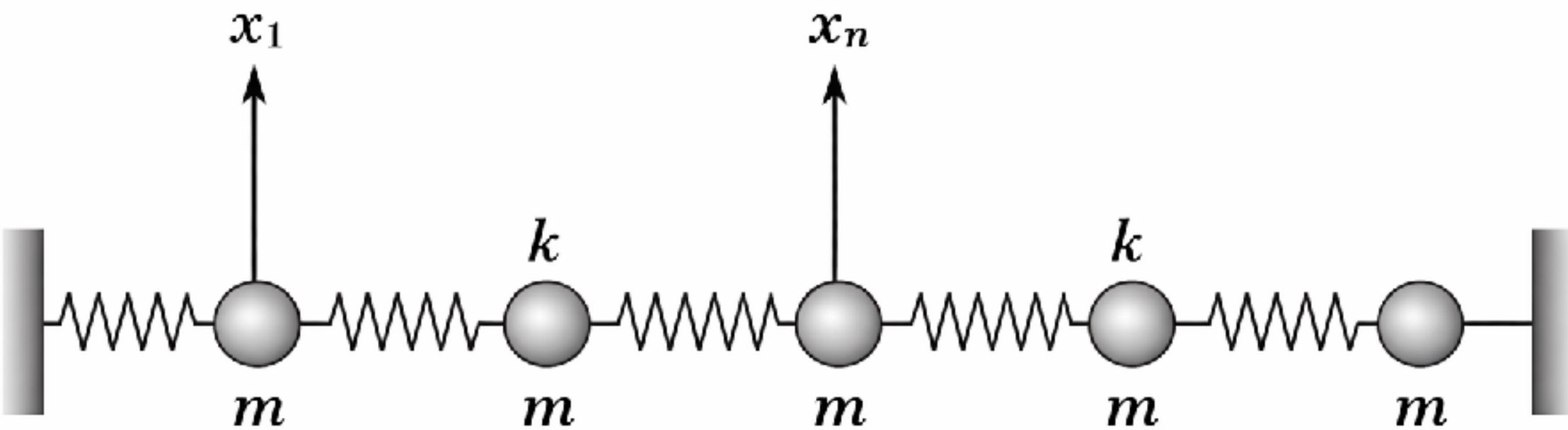
Stanislaw Ulam (1940s):

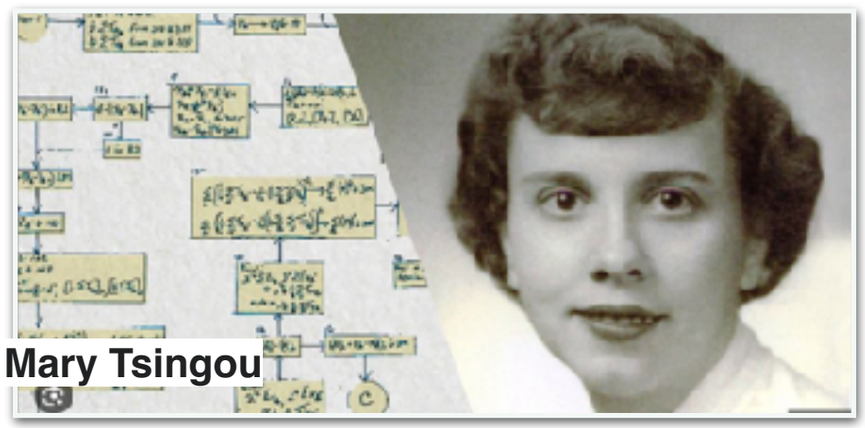
- Collaboration at Manhattan Project → Los Alamos
- The discrete grid intuition
- Proposal: discrete systems on regular lattice

Fermi-Pasta-Ulam (1953-1955): The first “computational experiment"

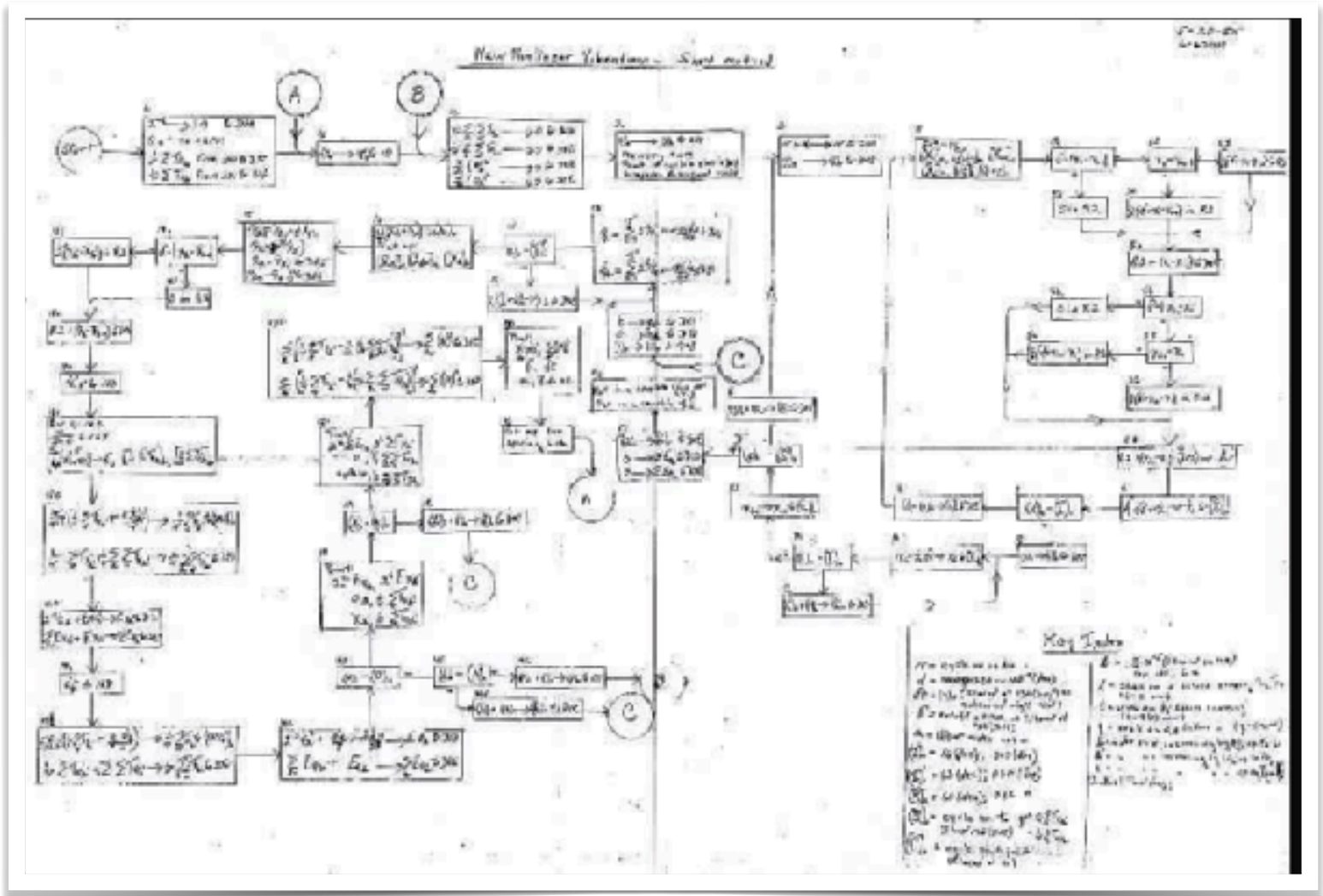


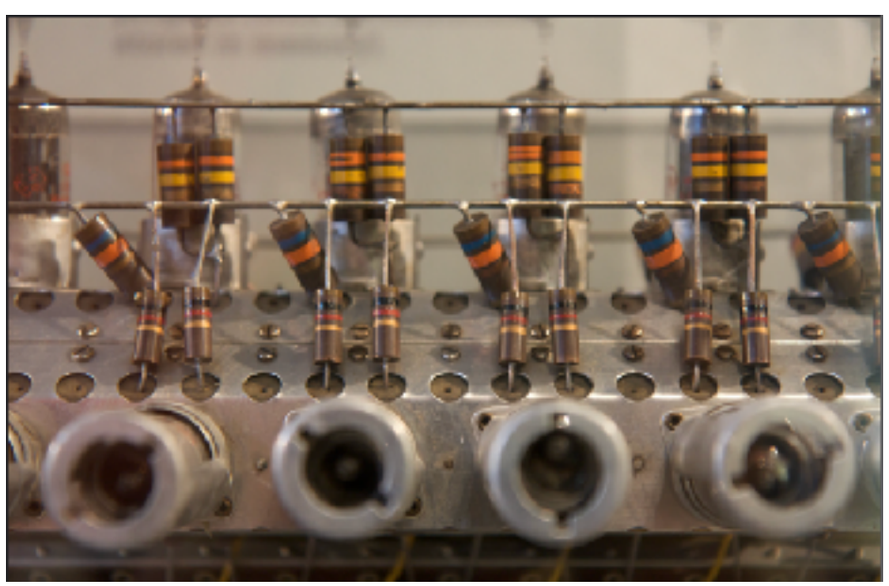
- **Fermi-Pasta-Ulam (1953-1955)**: The first "computational experiment"
  - ⚠️ **Note:** NOT a cellular automaton, but a chain of oscillators
  - Importance: first use of computer to study nonlinear dynamics
  - Surprising result: lack of thermalization

Today The TFPU Model (1953-1955):



Mary Tsingou

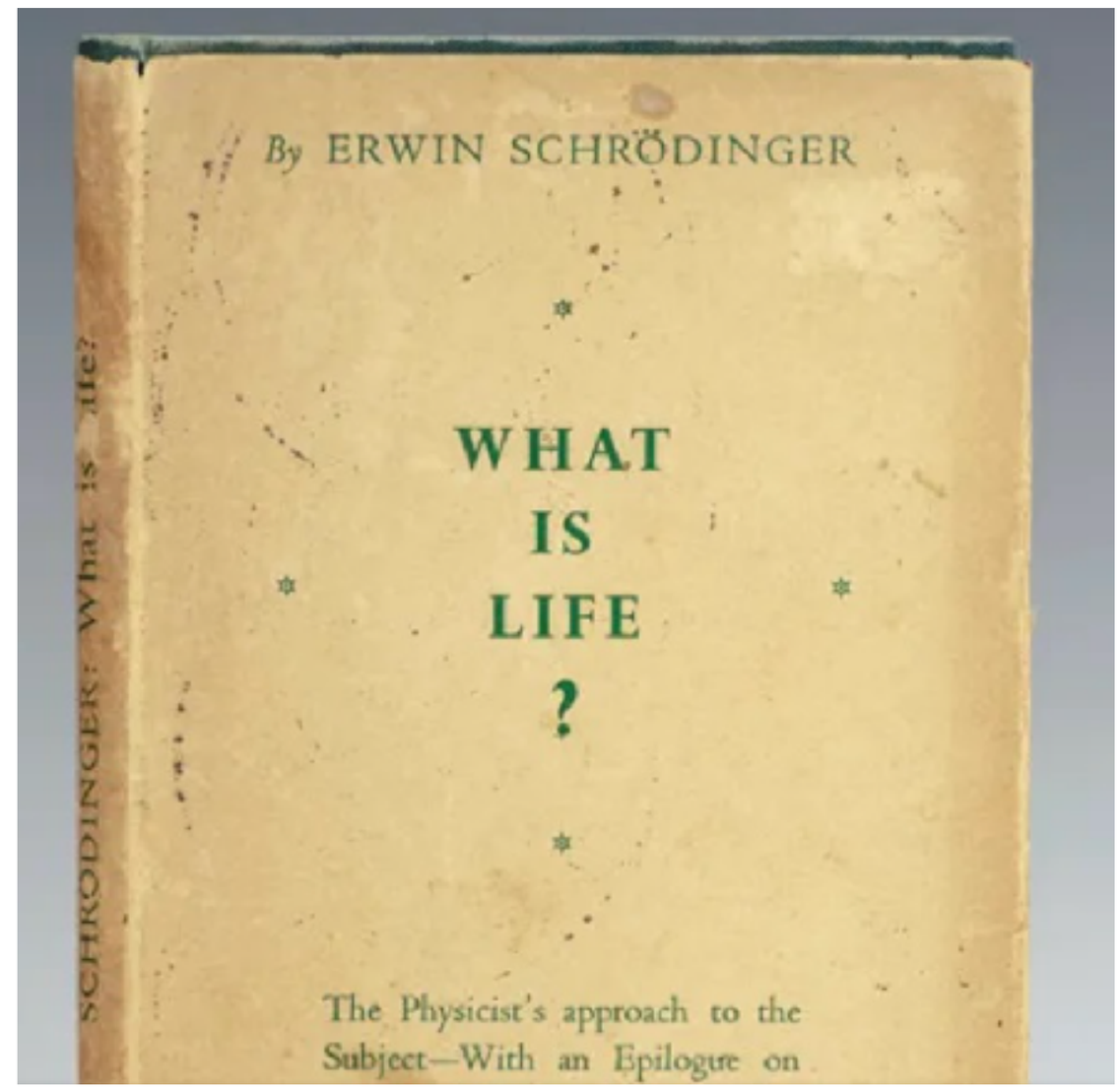




MANIAC tubes and circuits

To finish the Story……and CS and Genomics are born

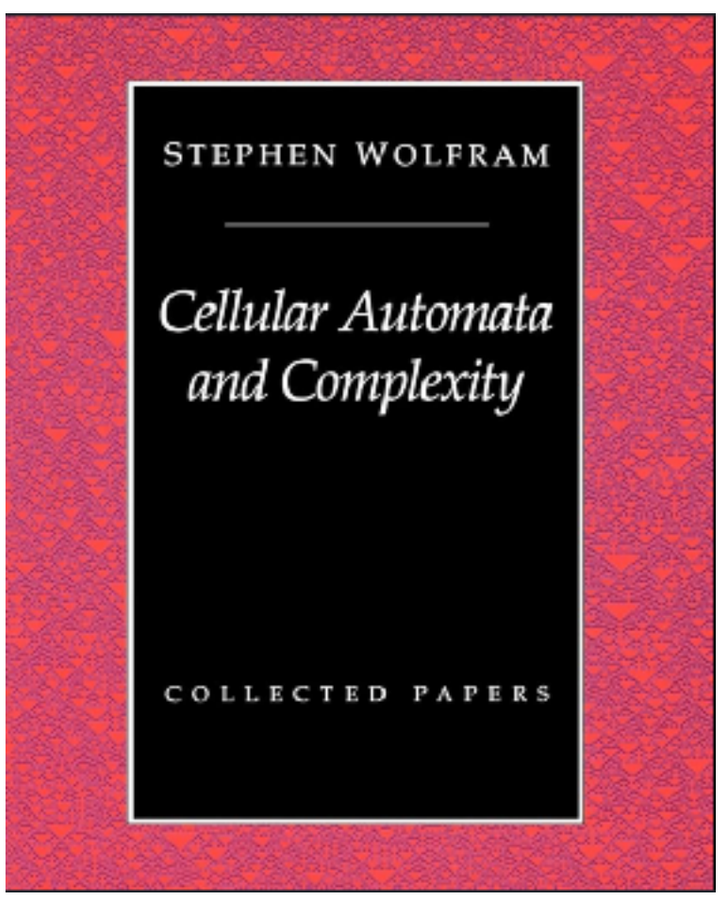
Erwin Schrödinger



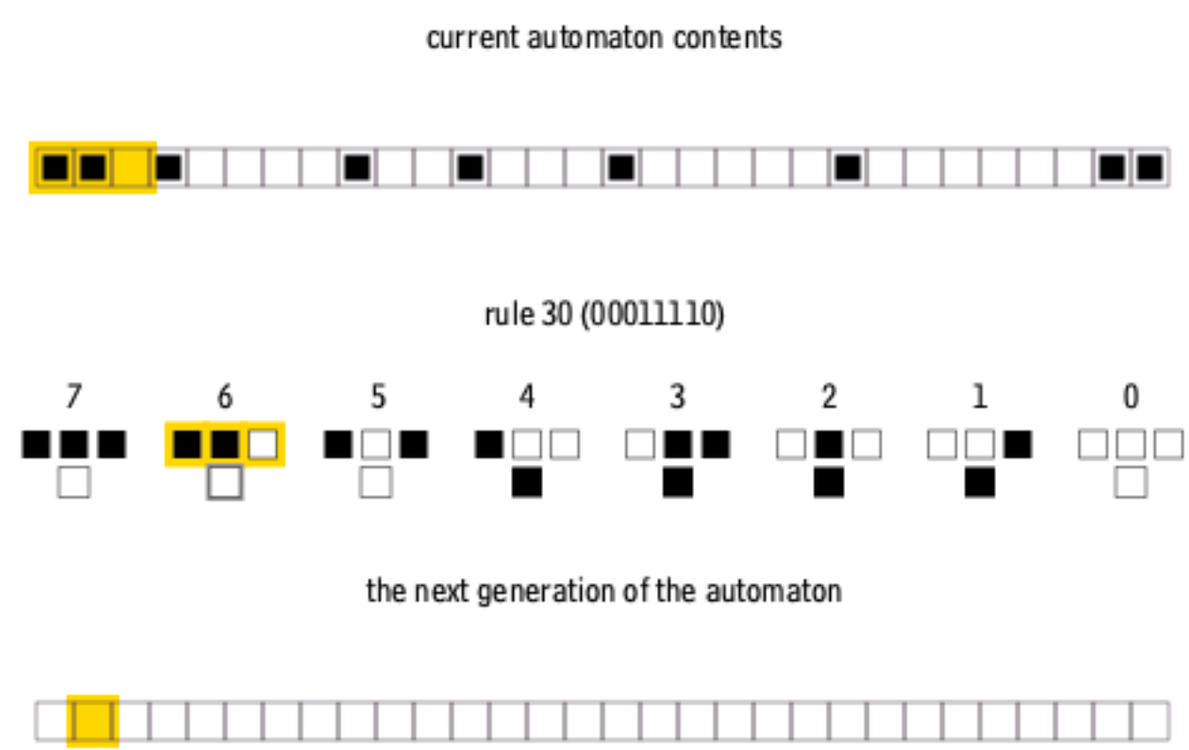
From complexity to simplicity

The crucial simplification

- Von Neumann + Ulam → cellular automata on 2D grid
- Problem: still very complex for systematic analysis
- **Wolfram (1980s)**: "What if we start from the simplest dimension?"
  - One spatial dimension
  - Two states (0/1, white/black)
  - Local deterministic rules
- Result: **unexpected behavioral richness**



GitHub



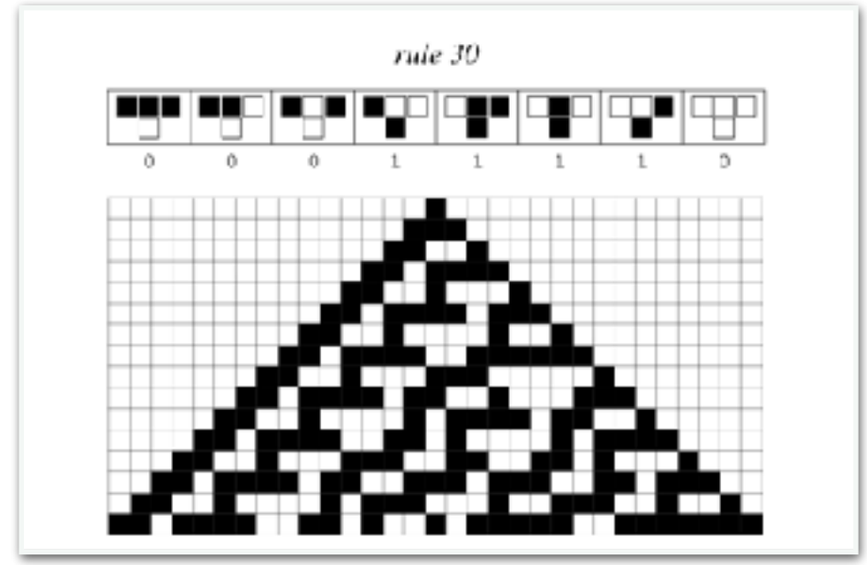
Rigorous definition of 1D CA

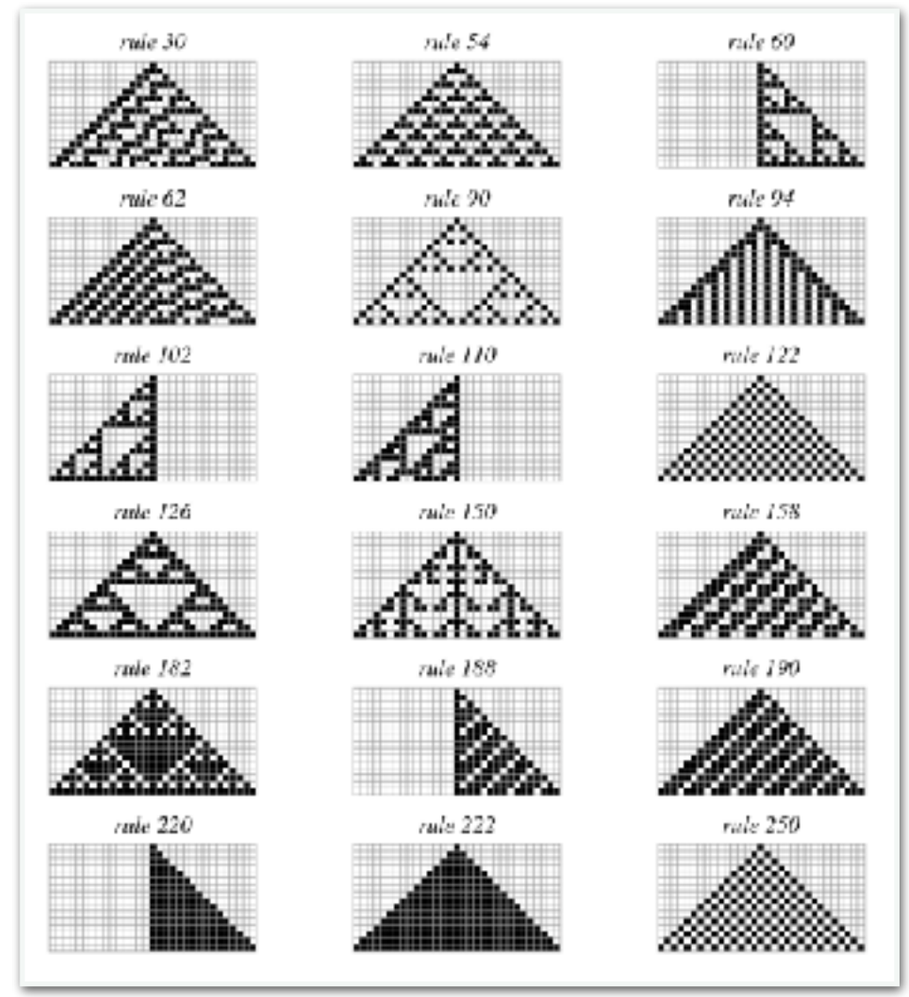
Fundamental components

A one-dimensional cellular automaton is defined by:

1. **Lattice:** $\mathbb{Z}$ (or $\mathbb{Z}_N$ for finite systems)
2. **States:** $S = \{0, 1\}$ (binary case)
3. **Configuration:** $c : \mathbb{Z} \to S$
   - At time t: $c^{(t)} = (...c_{-1}^{(t)}, c_0^{(t)}, c_1^{(t)}, ...)$
4. **Neighborhood:** r cells to left and right
   - $N_i = (c_{i-r}, ..., c_i, ..., c_{i+r})$
5. **Transition rule:** $\phi : S^{2r+1} \to S$
   - Deterministic, local, synchronous

Discrete dynamics: $c_i^{(t+1)} = \phi(c_{i-r}^{(t)}, ..., c_{i+r}^{(t)})$





Elementary Cellular Automata (ECA)

The simplest case: $r = 1$

- **Neighborhood:** 3 cells (left, center, right)
- Possible local configurations: $2^3 = 8$
- Possible rules: $2^8 = 256$

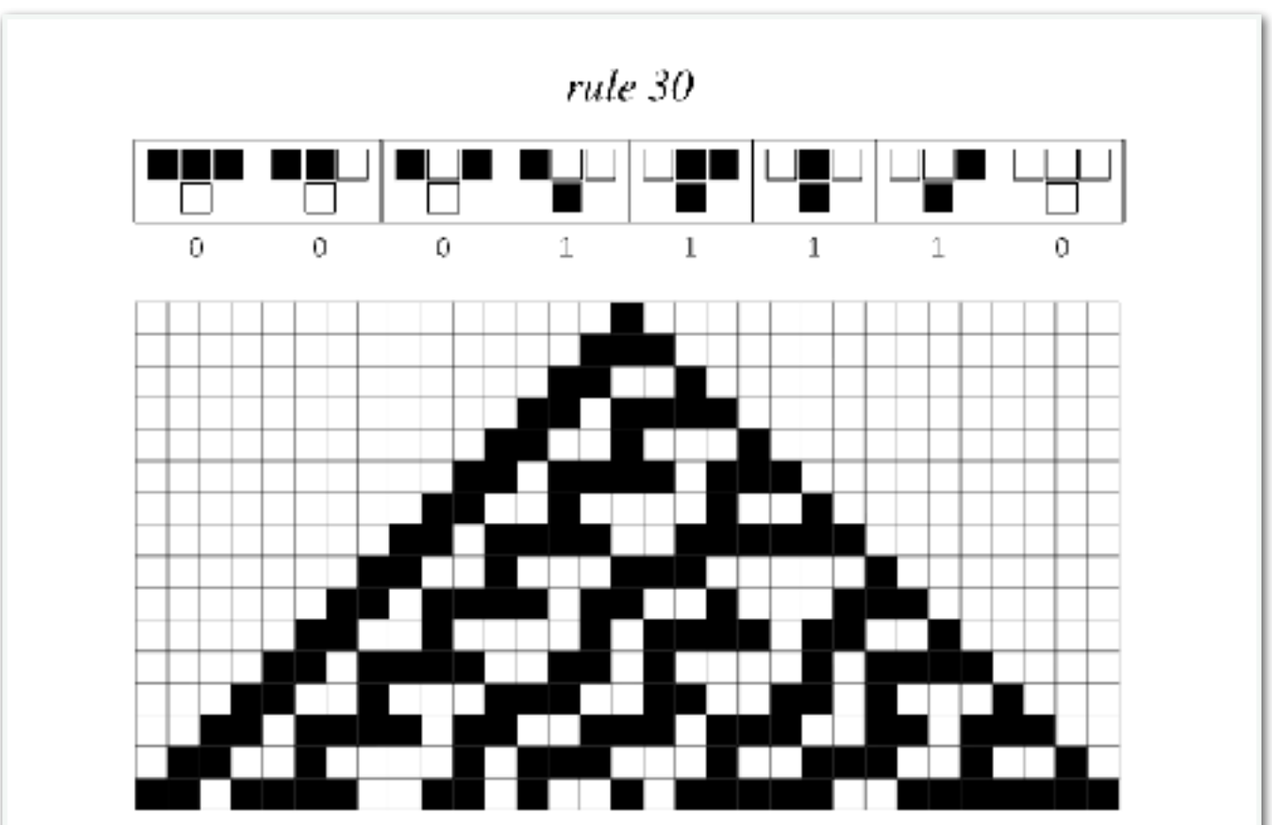
Wolfram encoding:

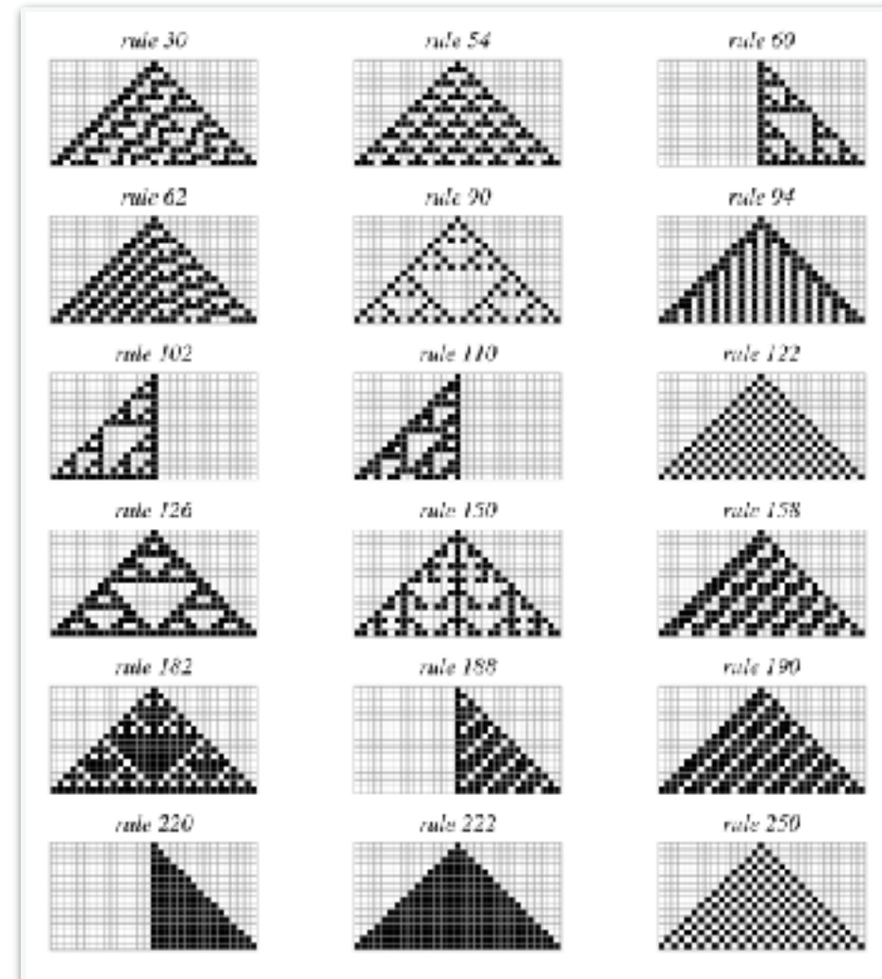
```
Neighborhood:  111  110  101  100  011  010  001  000
Output:         b₇   b₆   b₅   b₄   b₃   b₂   b₁   b₀
```

Rule number: $N = \sum_{i=0}^{7} b_i \cdot 2^i$

Example - Rule 30:

```
111→0, 110→0, 101→0, 100→1, 011→1, 010→1, 001→1, 000→0
Binary: 00011110₂ = 30₁₀
```



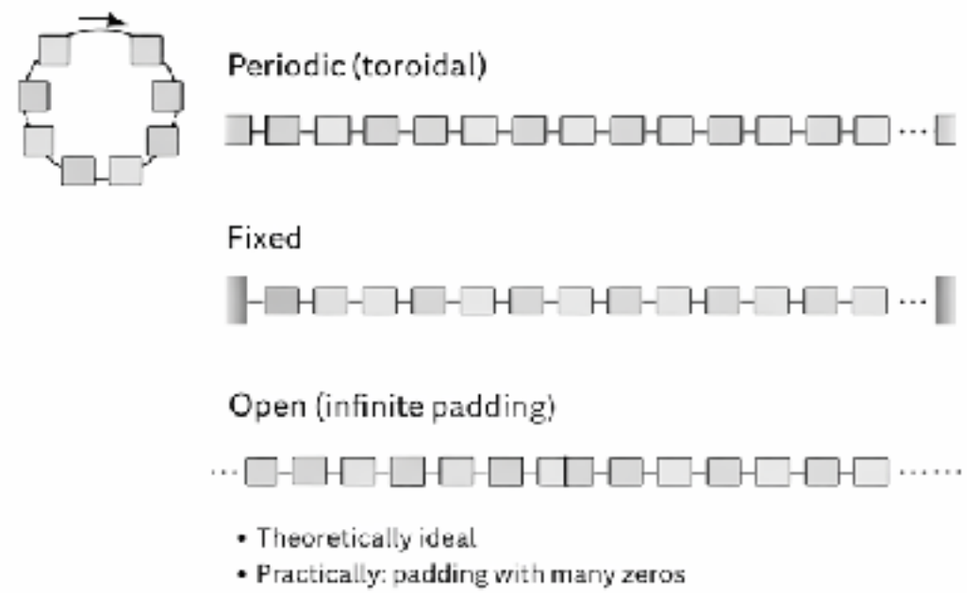


Boundary conditions

Finite systems: choices matter

1. **Periodic** (toroidal): $c_{i+N} = c_i$
   - Preserves translational invariance
   - No preferred "edge"
   - More natural for theoretical analysis
2. **Fixed:** $c_0 = c_{N-1} = 0$ (or 1)
   - Defined boundaries
   - Possible boundary effects
   - More natural for certain applications
3. **Open** (infinite padding): infinite lattice
   - Theoretically ideal
   - Practically: padding with many zeros

Trade-off: computational vs physical



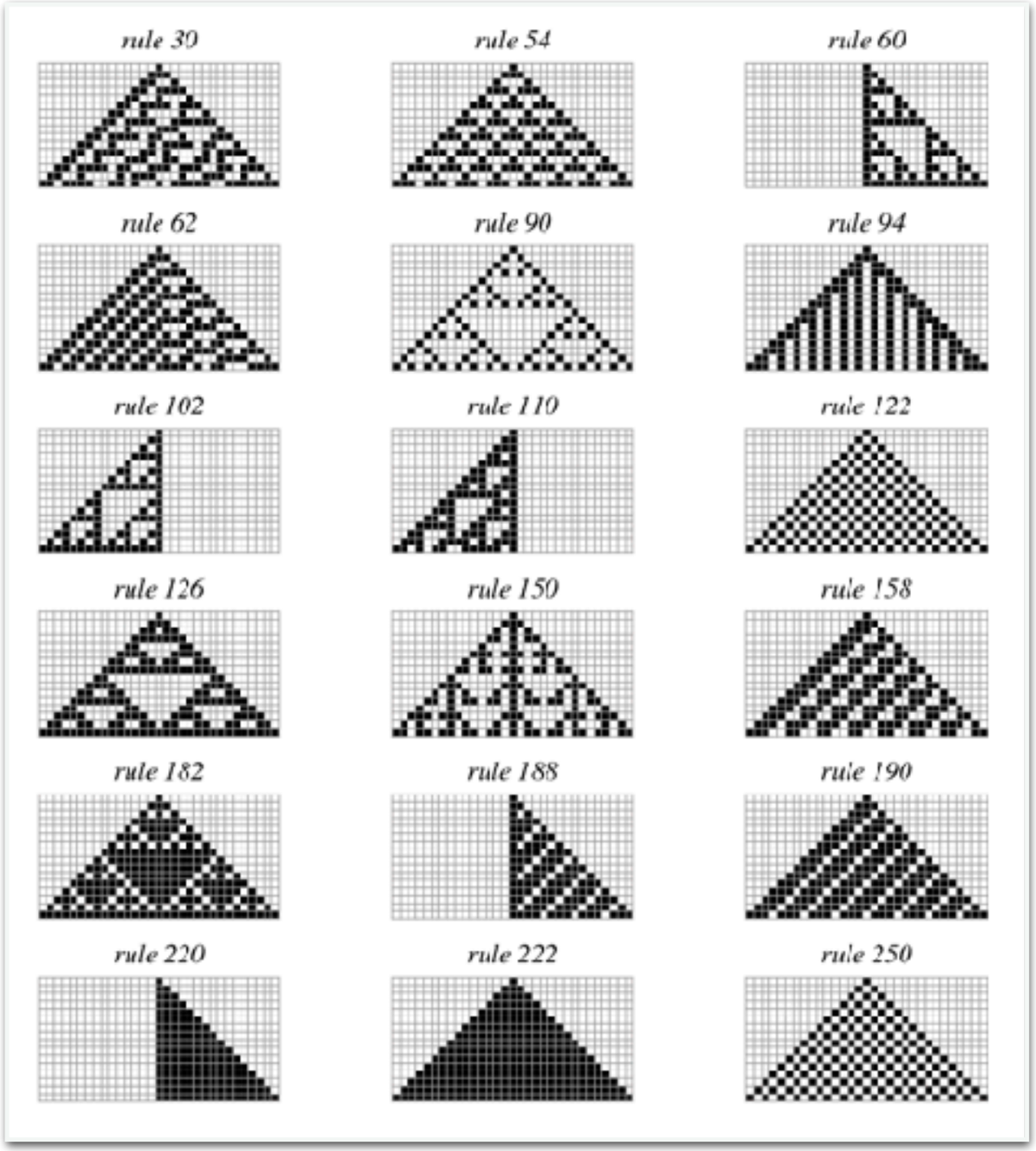
Space-time diagrams

The two-dimensional representation of time

- **Horizontal axis**: space (position i)
- **Vertical axis**: time (step t)
- **Color**: cell state (black=1, white=0)

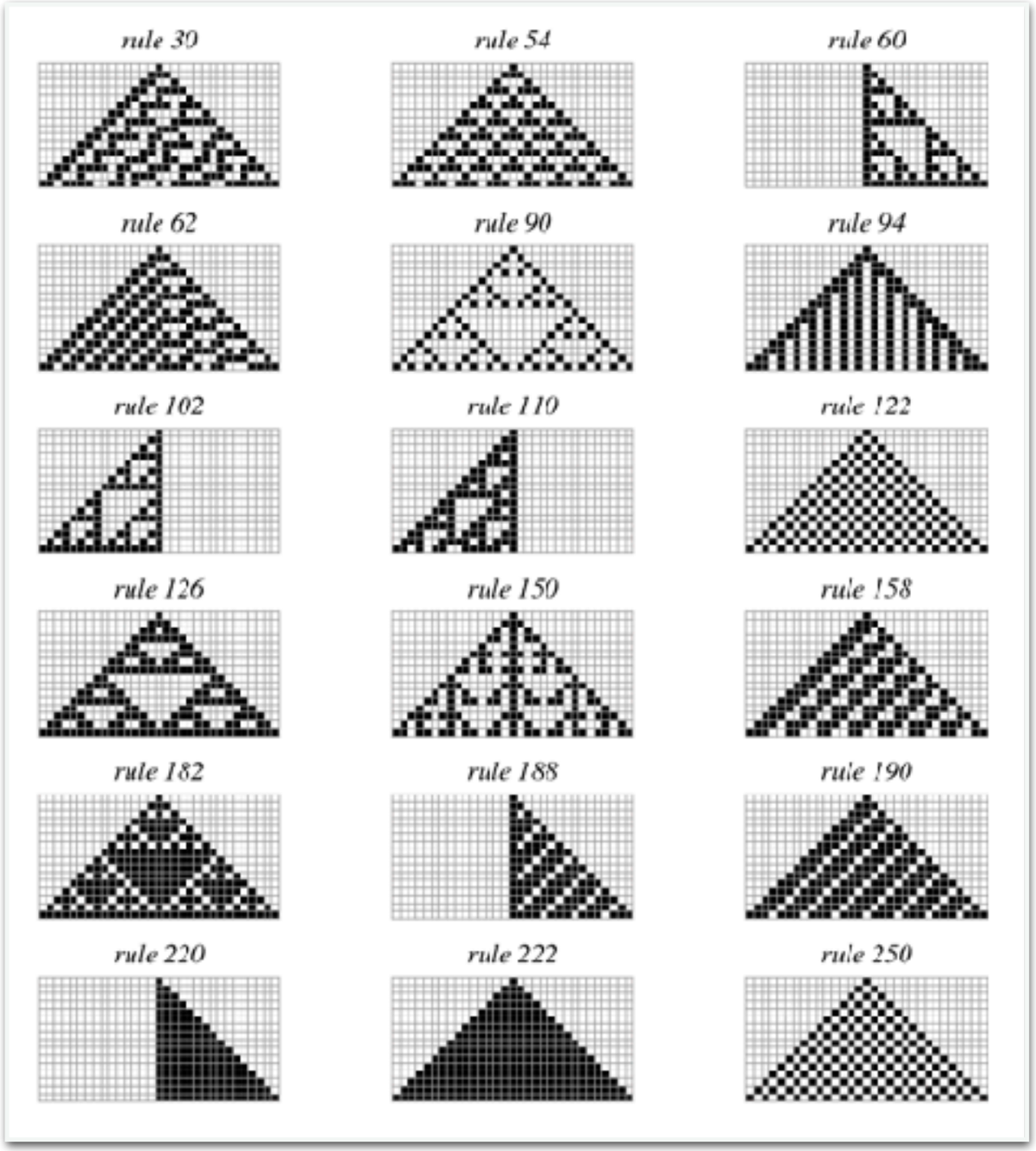
Advantages:

- Global patterns immediately visible
- Emergence of structures
- Behavioral classification



Iconic examples - First look

Rule	Behavior	Pattern
0	Uniform death	All white after t=1
254	All black	All black after t=1
90	Sierpiński	Fractal triangles
110	Complex	Localized structures



Wolfram's four classes (1984)

Qualitative classification of asymptotic behavior

Class I - Homogeneous

- Evolution toward uniform state (all 0 or all 1)
- Complete loss of information
- Examples: Rule 0, 32, 160, 250

Class II - Simple periodic

- Stable or oscillating periodic patterns
- Small periodicity (typically $\leqslant 4$)
- Examples: Rule 4, 108, 218

Class III - Chaotic

- Apparently random patterns
- Linear entropy growth
- Examples: Rule 30, 45, 73, 105

Class IV - Complex

- Persistent localized structures
- Non-trivial interactions
- Computationally rich
- Examples: Rule 110, 124

Rule 184 - Traffic flow model (class 2)

Physics application: particle transport

Physical interpretation:

- Particles move right if space is free
- Otherwise remain stationary
- Minimal traffic model!

Dynamics:

```
111→1, 110→1, 101→1, 100→0, 011→1, 010→0, 001→0, 000→0
```

System phases:

- $\rho < 0.5$: free flow (all move)
- $\rho = 0.5$: jamming transition
- $\rho > 0.5$: congested (many stationary)

Conservation: number of particles conserved!

Class III - Deterministic chaos

Chaotic yet deterministic

Characteristics:

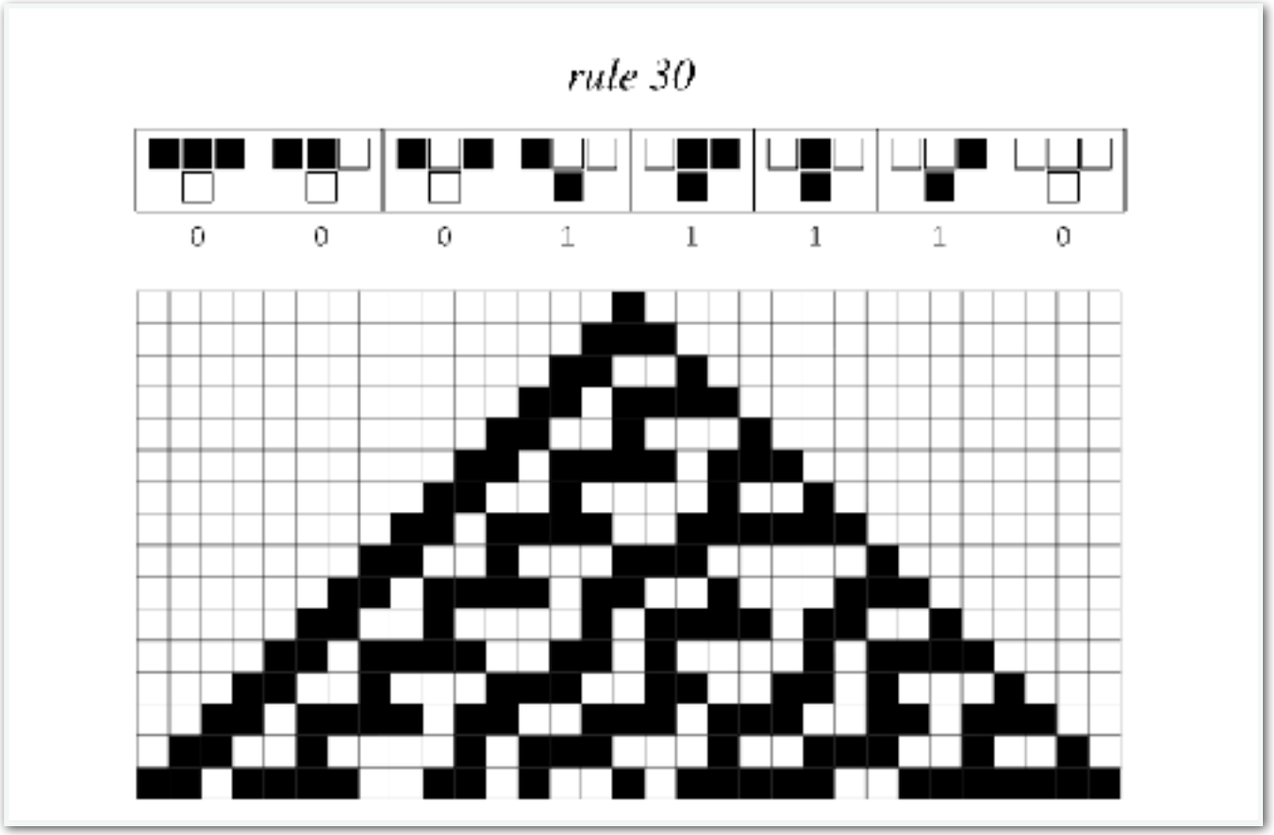
- Exponential sensitivity to initial conditions
- "Pseudo-random" patterns
- Linear complexity growth

Emblematic example - Rule 30:

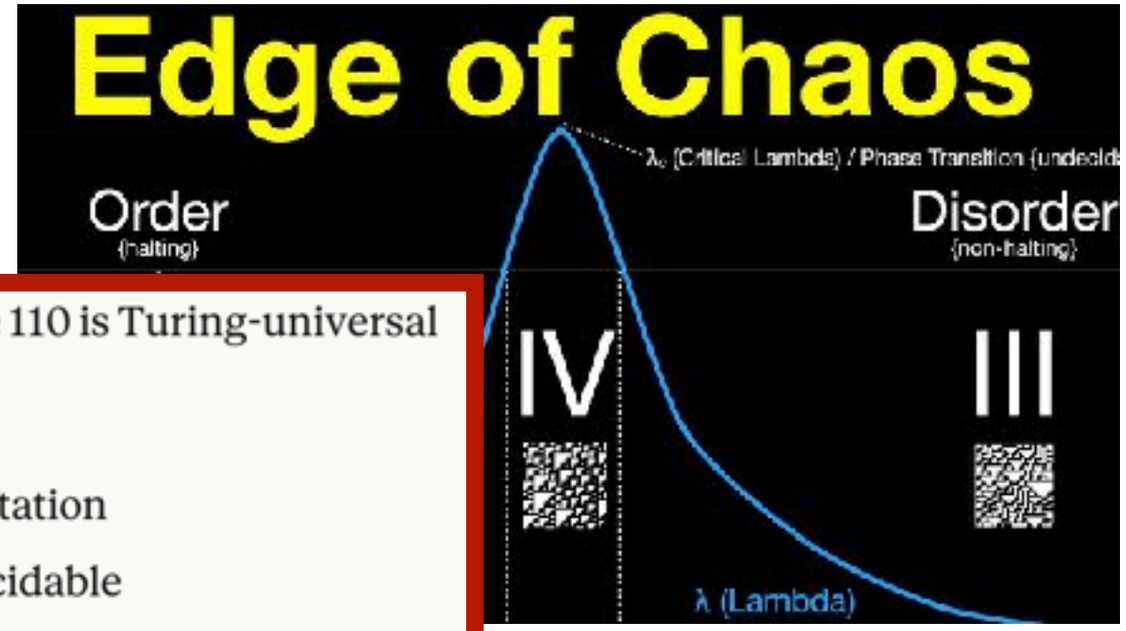
```
111→0, 110→0, 101→0, 100→1, 011→1, 010→1, 001→1, 000→0
```

Practical application:

- Wolfram uses Rule 30 as RNG in Mathematica
- Center column: high-quality pseudo-random sequence



Class IV - Computational complexity



Theorem (Cook, 2004): Rule 110 is Turing-universal

Implications:

- Can simulate any computation
- Halting problem is undecidable
- Intrinsic limit to predictability

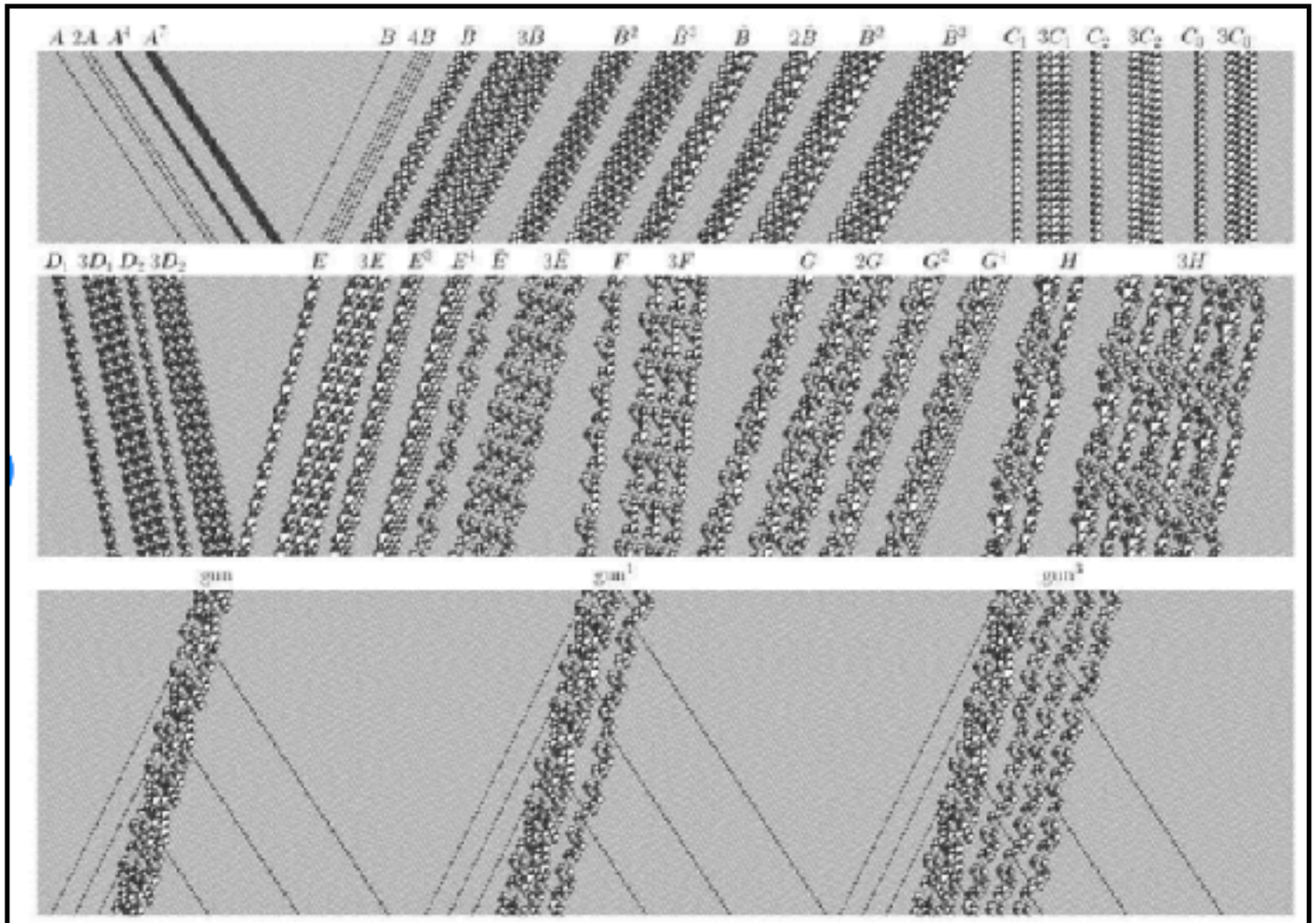
The edge of chaos

Unique characteristics:

- Localized structures ("gliders", "blinkers")
- Non-trivial interactions between structures
- Computational universality (Rule 110)
- "Edge of chaos" (Langton, Kauffman)

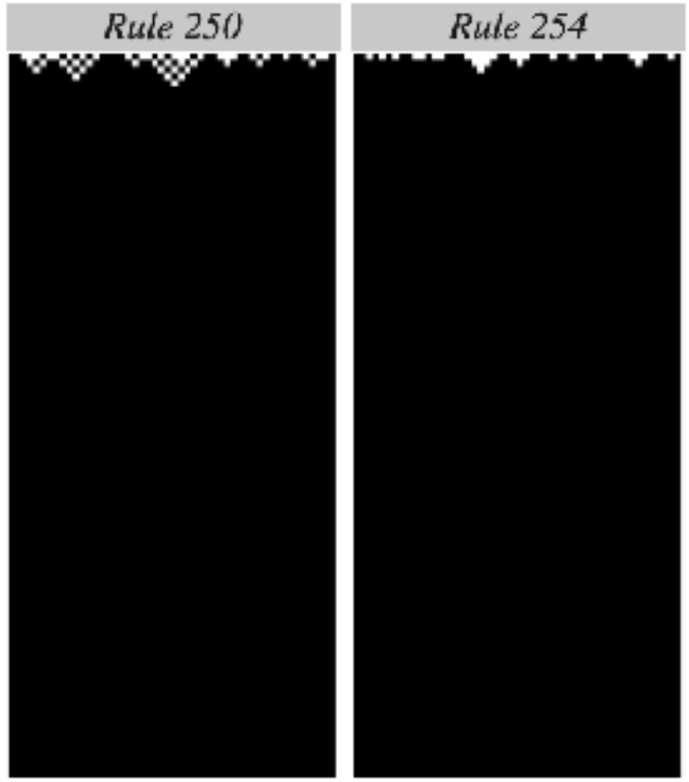
Example - Rule 110:

```
111→0, 110→1, 101→1, 100→0, 011→1, 010→1, 001→1, 000→0
```

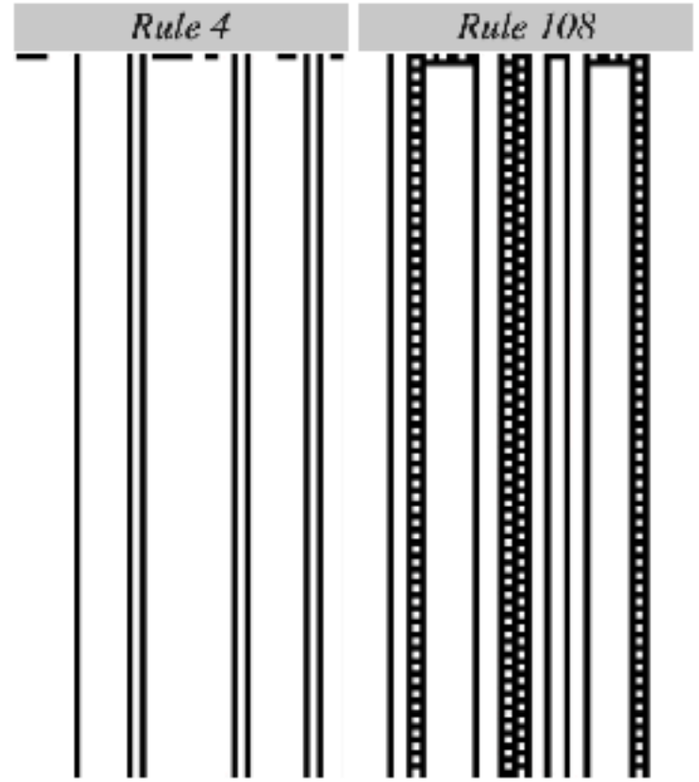


Glider classification in Rule 110.

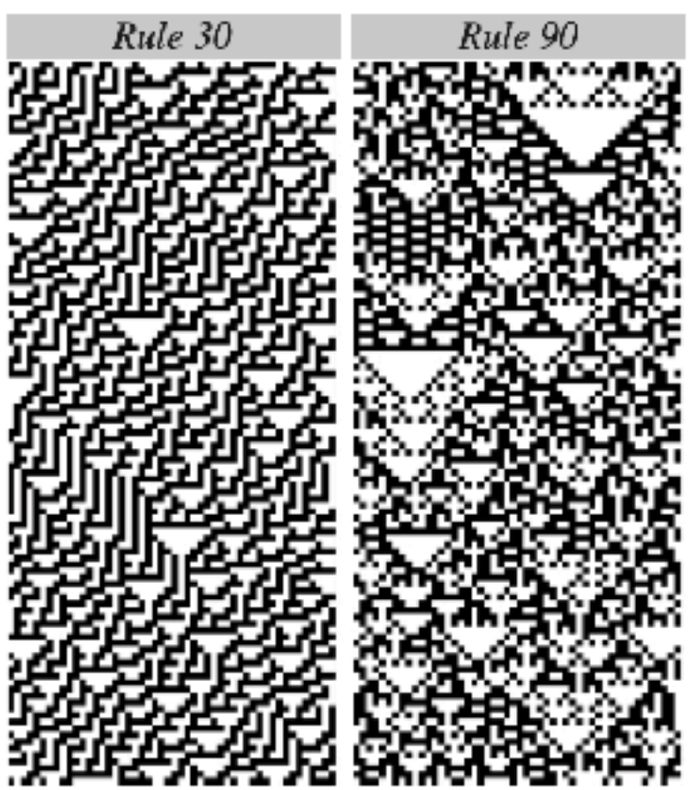
CLASS$_1$ rules leading to homogeneous states, all cells stably ending up with the same value:



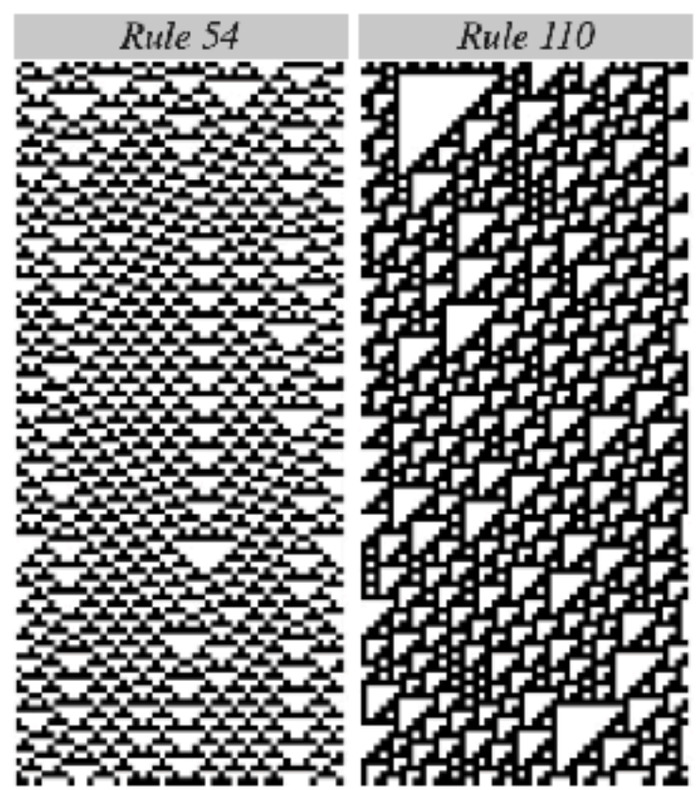
CLASS$_2$ rules leading to stable structures or simple periodic patterns:



CLASS$_3$ rules leading to seemingly chaotic, non-periodic behavior:



CLASS$_4$ rules leading to complex patterns and structures propagating locally in the lattice:

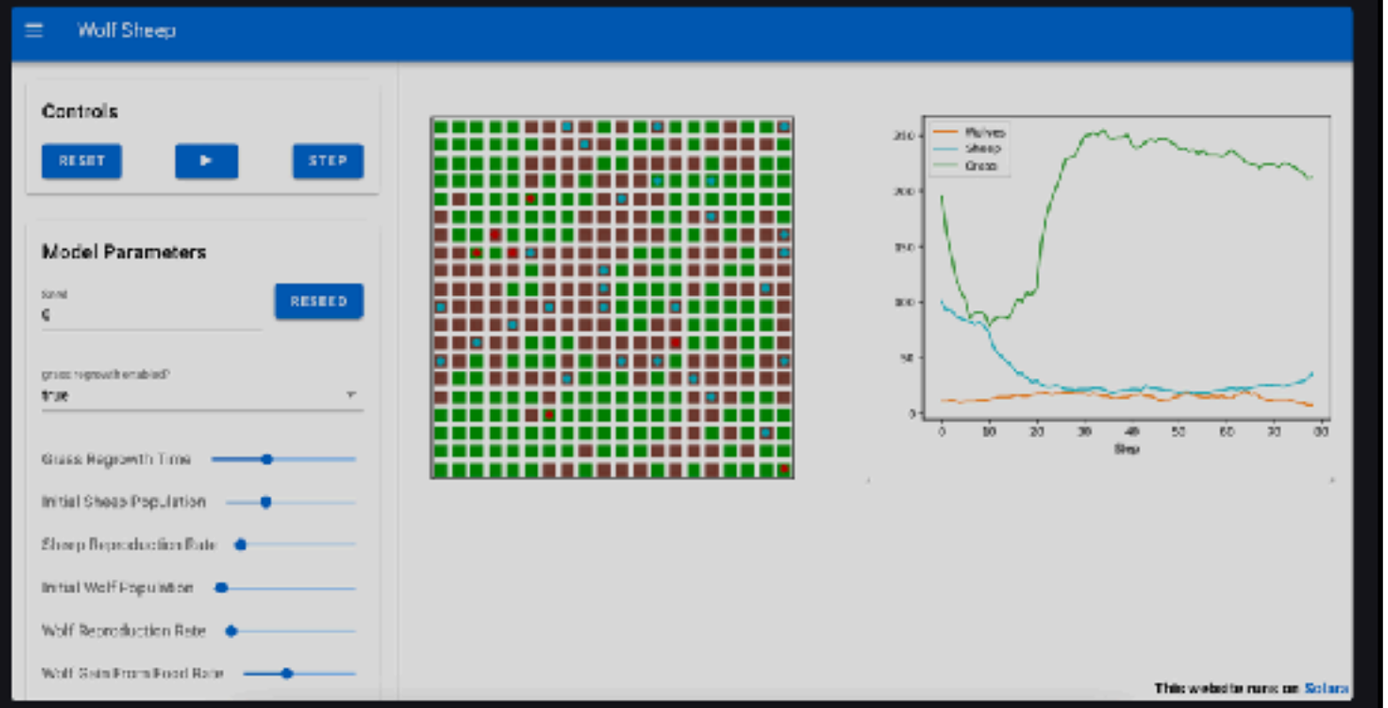


Mesa: Agent-based modeling in Python

JOSS 10.21105/joss.07668 | build passing | codecov 95% | code style black | chat 373 users

Mesa is an Apache2 licensed agent-based modeling (or ABM) framework in Python.

Mesa allows users to quickly create agent-based models using built-in core components (such as spatial grids and agent schedulers) or customized implementations; visualize them using a browser-based interface; and analyze their results using Python's data analysis tools. Mesa's goal is to make simulations accessible to everyone, so humanity can more effectively understand and solve complex problems.



A visualisation of the Wolf Sheep model build with Mesa. An online demo is available here.